AN ALGEBRAIC GROUP SCHEME OVER A FIELD IS QUASI-PROJECTIVE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma varieties-lemma-ample-after-field-extension (canonical tag 0BDC): ampleness descends from a field extension. cited at proof line 1588. Reduces to k algebraically closed.
- Proposition proposition-connected-component (canonical tag 0B7R): the connected component G^0 of an algebraic group scheme over a field is an open and closed normal subgroup scheme whose cosets are isomorphic to G^0. cited at proof lines 1590-1592. Identifies the non-identity components and cuts them out of the argument.
- Lemma lemma-group-scheme-characteristic-zero-smooth (canonical tag 047N) and Lemma lemma-reduced-group-scheme-prefect-field-characteristic-p-smooth (canonical tag 047P): smoothness in characteristic zero, resp. of the reduction in characteristic p; Lemma lemma-reduced-subgroup-scheme-perfect (canonical tag 047R) for H = G_red being a group scheme. cited at proof lines 1598-1612. Reduces to the smooth irreducible case.
- Divisors results: complement-of-affine-open is an effective Cartier divisor (0BCW), Weil divisors are Cartier over UFD local rings (0AGA), and the invertible sheaf of an effective Cartier divisor (01WX). cited at proof lines 1621-1671. Produce D on G and E on G x V and their invertible sheaves.
- Lemma morphisms-lemma-smooth-etale-over-affine-space (canonical tag 054L). cited at proof lines 1634-1637. Supplies the etale cover U -> A^d_k with the finite etale refinement W -> V.
- Lemma varieties-lemma-rational-equivalence-for-Pic (canonical tag 0BEH). cited at proof lines 1663-1665. Shows the class L of the fibre divisors is independent of v, so the sections s_v live in one invertible sheaf.
- Definition properties-definition-ample (canonical tag 01PS). cited at proof lines 1676-1678. The exhibited sections satisfy the defining property of ampleness.

Source reference. Original lemma 0BDC: sources/stacks-chow/tags.txt, source lines 14310–14310. Cited at proof line 1588 to reduce to the case of an algebraically closed ground field; load-bearing.

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Source reference. Original proposition 0B7R: sources/stacks-groupoids/pinned-original-source.tex, source lines 1230–1230. Cited at proof lines 1590-1592; the reduction to the identity component G^0 rests on it; load-bearing.

Source reference. Original lemma 047N: sources/stacks-groupoids/pinned-original-source.tex, source lines 1483–1483. Cited at proof lines 1598-1600 for smoothness of G in characteristic zero; load-bearing.

Source reference. Original proposition 0BD4: sources/stacks-chow/tags.txt, source lines 14302–14302. Cited at proof lines 1602-1604: it suffices to exhibit an ample invertible sheaf on the reduction H; load-bearing.

Source reference. Original lemma 0B42: sources/stacks-chow/tags.txt, source lines 13985–13985. Cited parenthetically at proof lines 1604-1607 to identify having an ample invertible sheaf with quasi-projectivity; navigation for the framing.

Source reference. Original lemma 047R: sources/stacks-groupoids/pinned-original-source.tex, source lines 1063–1063. Cited at proof lines 1609-1610 to know H = G_red is a group scheme; load-bearing.

Source reference. Original lemma 047P: sources/stacks-groupoids/pinned-original-source.tex, source lines 1512–1512. Cited at proof lines 1610-1612 for smoothness of H in characteristic p; load-bearing.

Source reference. Original lemma 056S: sources/stacks-chow/tags.txt, source lines 6720–6720. Cited at proof lines 1617-1619 to see the local rings of G are regular; load-bearing.

Source reference. Original lemma 0AG0: sources/stacks-chow/tags.txt, source lines 13178–13178. Cited at proof lines 1618-1620 to upgrade regular local rings to UFDs; load-bearing.

Source reference. Original lemma 0BCW: sources/stacks-chow/tags.txt, source lines 14294–14294. Cited at proof lines 1621-1626 to produce the Cartier divisor D with G \\ D affine; load-bearing.

Source reference. Original lemma 047L: sources/stacks-groupoids/pinned-original-source.tex, source lines 983–983. Cited parenthetically at proof lines 1624-1625; used when intersecting affine opens inside G; load-bearing.

Source reference. Original lemma 054L: sources/stacks-chow/tags.txt, source lines 6644–6644. Cited at proof lines 1634-1637 to choose the etale cover U -> A^d_k; load-bearing.

Source reference. Original section 0BCX: sources/stacks-chow/tags.txt, source lines 14295–14295. Cited inside the footnote at proof lines 1638-1641 as an alternative route to the divisor E; navigation for an alternative, not for the counted argument.

Source reference. Original lemma 0AGA: sources/stacks-chow/tags.txt, source lines 13188–13188. Cited at proof lines 1645-1649 to conclude the components of T are effective Cartier divisors; load-bearing.

Source reference. Original definition 01WX: sources/stacks-chow/tags.txt, source lines 2498–2498. Cited at proof line 1662 to define M = O(E) and at lines 1668-1671 for the canonical section; load-bearing.

Source reference. Original lemma 0BEH: sources/stacks-chow/tags.txt, source lines 14350–14350. Cited at proof lines 1663-1665 to see the isomorphism class L of M_v is independent of v; load-bearing.

Source reference. Original definition 01PS: sources/stacks-chow/tags.txt, source lines 2241–2241. Cited at proof lines 1676-1678 to conclude L is ample by definition; load-bearing.

Lemma 0.2. *Let k be a field. Let G be an algebraic group scheme over k. Then G is quasi-projective over k.*

Proof. By Varieties, Lemma 0BDC we may assume that k is algebraically closed. Let $G^0 \subset G$ be the connected component of G as in Proposition 0B7R. Then every other connected component of G has a k-rational point and hence is isomorphic to G^0 as a scheme. Since G is quasi-compact and Noetherian, there are finitely many of these connected components. Thus we reduce to the case discussed in the next paragraph.

Let G be a connected algebraic group scheme over an algebraically closed field k. If the characteristic of k is zero, then G is smooth over k by Lemma 047N. If the characteristic of k is $p > 0$, then we let $H = G_{red}$ be the reduction of G. By Divisors, Proposition 0BD4 it suffices to show that H has an ample invertible sheaf. (For an algebraic scheme over k having an ample invertible sheaf is equivalent to being quasi-projective over k, see for example the very general More on Morphisms, Lemma 0B42.) By Lemma 047R we see that H is a group scheme over k. By Lemma 047P we see that H is smooth over k. This reduces us to the situation discussed in the next paragraph.

Let G be a quasi-compact irreducible smooth group scheme over an algebraically closed field k. Observe that the local rings of G are regular and hence UFDs (Varieties, Lemma 056S and More on Algebra, Lemma 0AG0). The complement of a nonempty affine open of G is the support of an effective Cartier divisor D. This follows from Divisors, Lemma 0BCW. (Observe that G is separated by Lemma 047L.) We conclude there exists an effective Cartier divisor $D \subset G$ such that $G \setminus D$ is affine. We will use below that for any $n \geq 1$ and $g_1, \ldots, g_n \in G(k)$ the complement $G \setminus \bigcup Dg_i$ is affine. Namely, it is the intersection of the affine opens $G \setminus Dg_i \cong G \setminus D$ in the separated scheme G.

We may choose the top row of the diagram

$$\begin{array}{ccccc} G & \xleftarrow{j} & U & \xrightarrow{\pi} & \mathbf{A}^d_k \\ & & \uparrow & & \uparrow \\ & & W & \xrightarrow{\pi'} & V \end{array}$$

such that $U \neq \emptyset$, $j : U \to G$ is an open immersion, and π is étale, see Morphisms, Lemma 054L. There is a nonempty affine open $V \subset \mathbf{A}^d_k$ such that with $W = \pi^{-1}(V)$ the morphism $\pi' = \pi|_W : W \to V$ is finite étale. In particular π' is finite locally free, say of degree n. Consider the effective Cartier divisor

$$\mathcal{D} = \{(g, w) \mid m(g, j(w)) \in D\} \subset G \times W$$

(This is the restriction to $G \times W$ of the pullback of $D \subset G$ under the flat morphism $m : G \times G \to G$.) Consider the closed subset[1] $T = (1 \times \pi')(\mathcal{D}) \subset G \times V$. Since π' is finite locally free, every irreducible component of T has codimension 1 in $G \times V$. Since $G \times V$ is smooth over k we conclude these components are effective Cartier divisors (Divisors, Lemma 0AGA and lemmas cited above) and hence T is the support of an effective Cartier divisor E in $G \times V$. If $v \in V(k)$, then $(\pi')^{-1}(v) = \{w_1, \ldots, w_n\} \subset W(k)$ and we see that

$$E_v = \bigcup\nolimits_{i=1,\ldots,n} Dj(w_i)^{-1}$$

in G set theoretically. In particular we see that $G \setminus E_v$ is affine open (see above). Moreover, if $g \in G(k)$, then there exists a $v \in V$ such that $g \notin E_v$. Namely, the set W' of $w \in W$ such that $g \notin Dj(w)^{-1}$ is nonempty open and it suffices to pick v such that the fibre of $W' \to V$ over v has n elements.

Consider the invertible sheaf $\mathcal{M} = \mathcal{O}_{G \times V}(E)$ on $G \times V$. By Varieties, Lemma 0BEH the isomorphism class $\mathcal{L}$ of the restriction $\mathcal{M}_v = \mathcal{O}_G(E_v)$ is independent of $v \in V(k)$. On the other hand, for every $g \in G(k)$ we can find a v such that $g \notin E_v$ and such that $G \setminus E_v$ is affine. Thus the canonical section (Divisors, Definition 01WX) of $\mathcal{O}_G(E_v)$ corresponds to a section s_v of $\mathcal{L}$ which does not vanish at g and such that G_{s_v} is affine. This means that $\mathcal{L}$ is ample by definition (Properties, Definition 01PS). $\square$

[1] Using the material in Divisors, Section 0BCX we could take as effective Cartier divisor E the norm of the effective Cartier divisor $\mathcal{D}$ along the finite locally free morphism $1 \times \pi'$ bypassing some of the arguments.